

4.1 Orthogonality and the Four Fundamental Subspaces

Concise exam revision notes for orthogonal complements in $\mathbb{R}^n$ and $\mathbb{R}^m$.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Orthogonality connects the four fundamental subspaces in pairs. Row space is perpendicular to nullspace; column space is perpendicular to left nullspace.

The dot product test is

$$v^T w = 0.$$

If this holds for every vector v in one subspace and every vector w in another, the two subspaces are orthogonal.

2 Orthogonal Subspaces vs Orthogonal Complements

Orthogonal subspaces may intersect only at zero, but that alone does not mean they fill the whole space.

Idea	Meaning	Exam check
Orthogonal subspaces	Every vector in one is perpendicular to every vector in the other.	Dot products are zero.
Orthogonal complements	Orthogonal subspaces whose dimensions add to the ambient dimension.	Dot products zero and dimensions add correctly.

Common mistake. Do not say “orthogonal complement” from perpendicularity alone. Also check the dimension count.

3 Row Space and Nullspace

Let $A \in \mathbb{R}^{m \times n}$. Each row of A is a vector in $\mathbb{R}^n$. If $x \in N(A)$, then

$$Ax = 0.$$

This means every row of A has dot product zero with x . Therefore

$$C(A^T) \perp N(A).$$

Their dimensions add to the whole ambient space $\mathbb{R}^n$:

$$\dim C(A^T) + \dim N(A) = r + (n - r) = n.$$

So they are orthogonal complements:

$$N(A) = C(A^T)^\perp, \quad C(A^T) = N(A)^\perp.$$

4 Column Space and Left Nullspace

The left nullspace is

$$N(A^T) = \{y : A^T y = 0\}.$$

If $y \in N(A^T)$, then y is perpendicular to every column of A . Therefore

$$C(A) \perp N(A^T).$$

Their dimensions add to the whole ambient space $\mathbb{R}^m$:

$$\dim C(A) + \dim N(A^T) = r + (m - r) = m.$$

So they are orthogonal complements:

$$N(A^T) = C(A)^\perp, \quad C(A) = N(A^T)^\perp.$$

5 Four-Subspace Summary

Subspace	Lives in	Dimension	Orthogonal complement
Row space $C(A^T)$	$\mathbb{R}^n$	r	Nullspace $N(A)$.
Nullspace $N(A)$	$\mathbb{R}^n$	$n - r$	Row space $C(A^T)$.
Column space $C(A)$	$\mathbb{R}^m$	r	Left nullspace $N(A^T)$.
Left nullspace $N(A^T)$	$\mathbb{R}^m$	$m - r$	Column space $C(A)$.

Exam memory. The complement must live in the same ambient space: row with null in $\mathbb{R}^n$, column with left null in $\mathbb{R}^m$.

6 Small Example

Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The row space is

$$C(A^T) = \text{span}\{(1, 2, 0), (0, 0, 1)\}.$$

The nullspace is found from

$$x_1 + 2x_2 = 0, \quad x_3 = 0,$$

so

$$N(A) = \text{span}\{(-2, 1, 0)\}.$$

The dot products are zero:

$$(1, 2, 0) \cdot (-2, 1, 0) = 0, \quad (0, 0, 1) \cdot (-2, 1, 0) = 0.$$

This shows the row space direction and nullspace direction are perpendicular in $\mathbb{R}^3$.

7 How to Use This in Problems

If you know a vector is perpendicular to	Then place it in
Every row of A	$N(A)$.
Every vector in $N(A)$	$C(A^T)$.
Every column of A	$N(A^T)$.
Every vector in $N(A^T)$	$C(A)$.

Dimension check. Perpendicularity plus the correct dimension identifies the whole orthogonal complement.

8 Common Mistakes

Mistake	Correct exam habit
Putting row space in $\mathbb{R}^m$.	Row vectors have length n , so row space lives in $\mathbb{R}^n$.
Pairing column space with $N(A)$.	Column space pairs with $N(A^T)$.
Forgetting the left nullspace.	Solve $A^T y = 0$, not $Ax = 0$.
Calling any perpendicular subspaces complements.	Check that the dimensions add to n or m .
Copying the picture instead of using the dimension count.	Use the table and formulas to reconstruct the relationship.

9 Final Exam Checklist

Checklist.

1. Write the size: A is $m \times n$.
2. Identify the ambient space: $\mathbb{R}^n$ or $\mathbb{R}^m$.
3. Pair $C(A^T)$ with $N(A)$ inside $\mathbb{R}^n$.
4. Pair $C(A)$ with $N(A^T)$ inside $\mathbb{R}^m$.
5. Use $Ax = 0$ for the nullspace.
6. Use $A^T y = 0$ for the left nullspace.
7. Check dimension sums: $r + (n - r) = n$ and $r + (m - r) = m$.
8. State the orthogonal complement relationship directly.